Soil Chemistry: Salinity Distribution

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This R Markdown file details creating a distribution plot using ggplot2 from a csv file. In particular, this script focuses on the environmental salinity distribution of various soil samples. For more information on basic density charts with ggplot2, reference the following link:
https://r-graph-gallery.com/21-distribution-plot-using-ggplot2.html

The following two web pages were referenced in conversion of salinity percentages to dS/m:
https://www.landscape.sa.gov.au/mr/publications/measuring-salinity
https://watereuse.org/salinity-management/ls/ls_3d.html

Start by loading in the ggplot2 package.

```
library(ggplot2)
```

Next, read in the csv file as the function. The as.numeric() function allows one to convert strings to numerical values for plotting.

```
density_function <- read.csv("CIB_Soil_Chemistry_Salinity.csv")
density_function$Salinity_dS_m <- as.numeric(density_function$Salinity_dS_m)
```

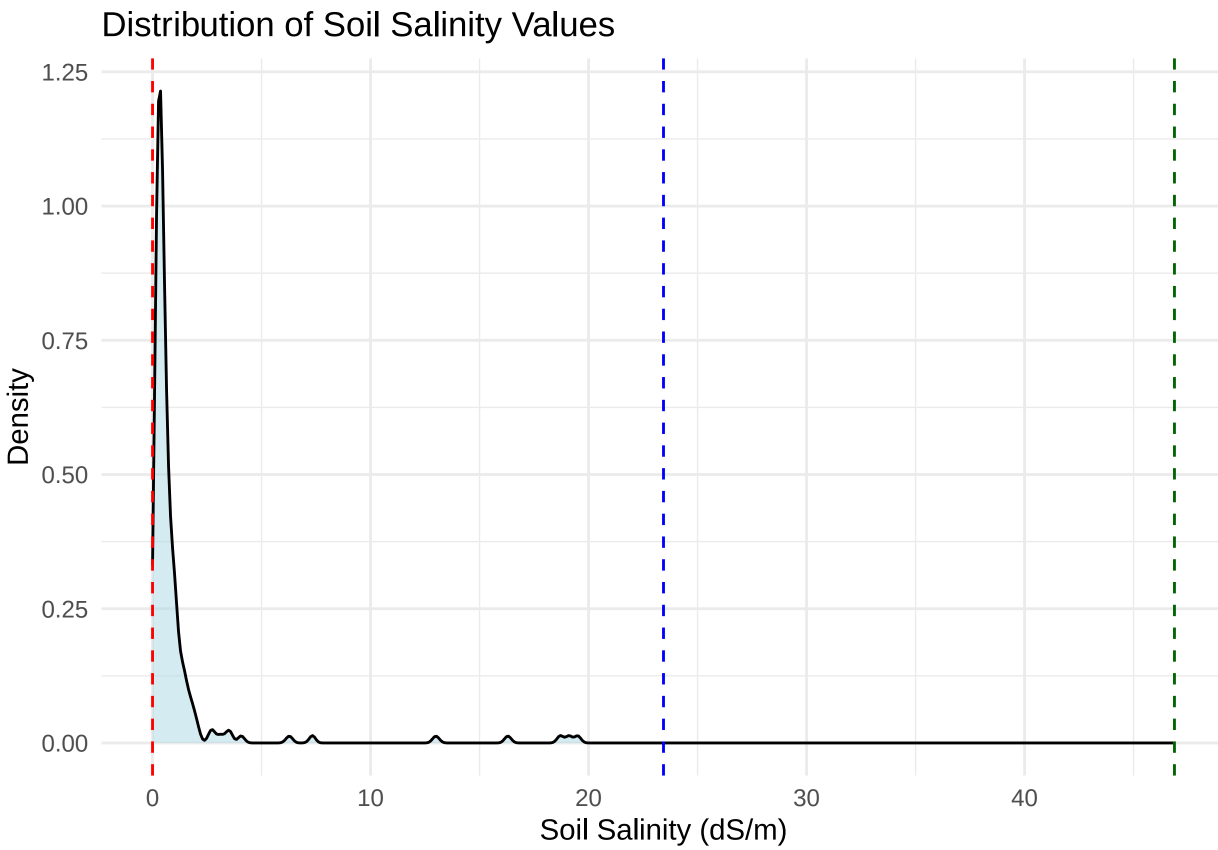
Finally, set up the script for running the plot. In this case, vertical lines were introduced to demonstrate experimental media salinity values for evaluation against initial environmental conditions.

```
ggplot(density_function, aes(x = Salinity_dS_m)) +
  geom_density(fill = "lightblue", alpha = 0.5) +

  # create vertical lines
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 23.4375, linetype = "dashed", color = "blue") +
  geom_vline(xintercept = 46.875, linetype = "dashed", color = "darkgreen") +

  theme_minimal() +

  # create labels
  labs(title = "Distribution of Soil Salinity Values", x = "Soil Salinity (dS/m)",
       y = "Density"
  )
```


Distribution of Soil Salinity Values
Density
1.25
1.00
0.75
0.50
0.25
0.00
0
10
20
30
40
Soil Salinity (dS/m)